

California Regional Water Quality Control Board
North Coast Region

REVISED MONITORING AND REPORTING PROGRAM NO. 89-2**
(Revised April 6, 1990)

FOR

CALIFORNIA DEPARTMENT OF CORRECTIONS
PELICAN BAY PRISON

Del Norte County

MONITORING

WASTEWATER TREATMENT FACILITY:

The purposes of monitoring the wastewater treatment facility are:

- To ensure adequate treatment of the wastewater including removal of suspended and floatable material, treatment of biodegradable organics, and elimination of pathogenic organisms.
- To routinely sample and analyze the treatment process; and
- To produce a sequence of reports which document the discharger's consistent compliance regarding adequate treatment, operation and maintenance.

Composite samples may be taken by a proportional sampling device approved by the Executive Officer or by grab samples composited in proportion to flow. In compositing grab samples, the sampling interval shall not exceed one hour. The following shall constitute the monitoring program:

Influent Monitoring

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
BOD (20°C, 5-day)	mg/l	8-hour composite	twice weekly
Total Phosphorous	mg/l	8-hour composite	monthly
Nonfilterable Residue	mg/l	8-hour composite	daily
Flow (Mean and Peak)	mgd	continuous	daily

Effluent Monitoring

Samples are to be taken after disinfection.

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Mean Daily Flow	mgd	continuous	daily
BOD (20°C, 5-day)	mg/l	8-hour composite	twice weekly
Settleable Matter	ml/l	grab	daily

** Supercedes Monitoring and Reporting Program No. 89-2, dated January 26, 1989

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
\Nonfilterable Residue	mg/l	8-hour composite	daily
\Grease and Oil	mg/l	grab	monthly
\Hydrogen Ion	pH	grab	daily
\Coliform Organisms	MPN/100 ml	grab	daily
\Chlorine Residual	mg/l	grab	daily
Turbidity	NTU	continuous	daily
Nitrate Nitrogen	mg/l	8-hour composite	twice weekly
Ammonia Nitrogen	mg/l	8-hour composite	twice weekly
Total Kjeldahl Nitrogen	mg/l	8-hour composite	twice weekly
Organic Nitrogen	mg/l	8-hour composite	twice weekly
Total, Organic and Reactive Phosphorus	mg/l	8-hour composite	twice weekly
Chloride	mg/l	8-hour composite	twice weekly
Halogenated Volatile Organics (EPA Method 601)	mg/l	grab	quarterly

GROUNDWATER MONITORING:

Groundwater monitoring is conducted to:

- Identify water quality parameters which may be used to detect and/or quantify impacts of sludge application practices or treated wastewater discharge to the Rapid Infiltration Basins (RIBs);
- Describe water quality in groundwater prior to and after application of municipal sludge and discharge of treated wastewater to the RIBs;
- Obtain routine sampling and analysis information of groundwater quality conditions;
- Produce a sequence of reports which reveal whether or not impacts to groundwater quality have occurred due to sludge application or discharge of treated wastewater to RIBs.

Pre-Discharge Monitoring

Describe existing groundwater flow gradients. Install and maintain a network of piezometers within and proximate to the sludge disposal area, including at least one "background" monitoring well located in an area unaffected by the commencement of discharge.

Describe groundwater quality conditions. Install and maintain a network of sampling wells within and proximate to the sludge disposal area, sample and analyze the spatial and temporal variations of existing groundwater quality using the following parameters:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Total Dissolved Solids	mg/l	grab	monthly
Nitrate Nitrogen	mg/l	grab	monthly
Orthophosphate	mg/l	grab	monthly
Chloride	mg/l	grab	monthly

Propose a network of sampling/monitoring wells to serve as checkpoints for compliance with this Order.

Discharge Monitoring

Samples shall be taken from an established network of groundwater monitoring wells representative of conditions upgradient and downgradient of the sludge disposal area and rapid infiltration basins. The locations and construction details of the monitoring wells shall be in accordance with specifications issued by the Executive Officer.

Samples shall be analyzed for the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Total Dissolved Solids	mg/l	grab	monthly
Nitrate Nitrogen	mg/l	grab	monthly
Orthophosphate	mg/l	grab	monthly
Chloride	mg/l	grab	monthly

Continue to maintain the established network of piezometers; observe, record and analyze the variations in piezometric surface; the frequency shall coincide with any storm event which exceeds 2 inches of precipitation; monitoring shall include Smith River stage-height readings taken concurrently with the piezometer recordings; compare and analyze data obtained prior to discharge with data obtained after commencement of discharge.

SLUDGE DISPOSAL AREA:

Pre-discharge Monitoring

A representative sample of soil from the sludge application area shall be collected for metals (ICAP Scan). Following soil sample collection, a report shall be completed that describes the procedures and methods used to collect the soil sample and the laboratory analysis conducted on the sample. The report shall be submitted to the Regional Board before commencement of sludge disposal (injection).

Discharge Monitoring

Annually, following the completion of sludge disposal for the year, the discharger shall provide a report to the Board which shall include but not be limited to the following:

- The quantity of sludge disposed (in dry tons and gallons);
- The quality of sludge disposed (ie., well stabilized, odor free) and concentrations of nitrogen and the metals identified and quantified from the quarterly samples.

- c. The specific location on the application site where the sludge was injected or spread.
- d. Analyses of the soil pH prior to sludge disposal.
- e. The amount of nitrogen (N) in pounds applied per acre.

To assess the amount of nutrients removed by cropping, the discharger shall analyze a representative sample of the harvested crop for Total Nitrogen and Total Phosphorus. The crop analysis shall be used to assess the relative contribution of nutrients provided by sludge.

This report shall be submitted to the Regional Board and EPA no later than December 31st of each year.

Following the fourth year of sludge disposal, a representative sample of soil shall be collected from the application area and analyzed for the metals encountered during the quarterly sludge analyses. The results shall be included in the above report.

SMITH RIVER MONITORING:

The objectives of Smith River monitoring are to:

Describe ambient water quality conditions in the river prior to, and concurrent with the start of treated wastewater discharge to the Rapid Infiltration Basins (RIBs);

Identify water quality parameters which may be used to detect and/or quantify impacts of treated wastewater discharge;

Perform routine sampling and analysis of water quality conditions;

Produce a sequence of reports which reveal whether or not impacts of discharge have been detected.

Periodic Aerial Survey:

During the 1990 discharge monitoring period and at subsequent time periods as specified by the Regional Board, the discharger shall take still photographs to document the presence/absence and extent of algal accumulation at shorelines along lower Smith River. The five surface water monitoring stations as well as other strategic locations designated by the Regional Board shall be recorded. The photographs shall cover the water surface and adjacent gravel bars and be directed to document the presence/absence and/or extent of algae accumulations along the lower eight-mile reach of the river.

Summer-season Surface Water Monitoring:

The following analysis series shall be performed at the stations indicated:

Upper Station. Proximate to the Crescent City Water Intake.

Water Column Series¹

Periphyton Series²

Continuous Multiparameter Survey³

- 1/ Water Column Series consists of analyses of samples taken from well-mixed section of the river (except at the Below Rowdy Creek station, where sample series shall be obtained at two depths: above the pycnocline and below the pycnocline) at the locations indicated and analyzed for the following:

Field Parameters

Temperature
Electrical Conductivity
pH
Dissolved Oxygen

Lab Parameters

Turbidity
Nitrate-N
Total Kjeldahl-N
Total reactive phosphorus
Chloride
Total Iron
Biomass (chlorophyll a / adenosine triphosphate)

The frequency shall be monthly during the low flow period of the year, June through the beginning of winter runoff.

- 2/ Periphyton Series consists of descriptive (identification to genus and ranking as to relative abundance) and quantitative (dry weight and ash-free dry weight per unit of surface) information derived after deploying and retrieving standard plexiglass substrates as described by the following:

USGS, 1977. Methods for collection and analysis of aquatic biological and microbiological samples. Chapter A4 (eds.: Greeson, et al). Techniques of Water-Resources Investigation of the United State Geological Survey. Book 5. Washington DC.

USGS, 1979. A supplement to: Methods for collection and analysis of aquatic biological and microbiological samples (U.S. Geological Survey Techniques of Water-Resources Investigations, Book 5, Chapter A4). U.S. Geological Survey. Open-File Report 79-1279. Washington, DC.

The discharger shall deploy at least five plexiglass substrates per station and camouflage each to help prevent vandalism. The period of substrate deployment shall bracket the dates of Water Column sampling and the period during which Continuous Multiparameter Survey is conducted. The start of this series may be delayed beyond June if cold temperatures or high flows persist.

- 3/ Continuous Multiparameter Survey consists of sequential recording of Temperature, pH, Electrical Conductivity and Dissolved Oxygen via an in-situ probe which is deployed in the river for at least a 48-hour period. The time of deployment shall include the date that Water Column sampling is done and shall "nest" within the period that Periphyton substrate is deployed.

Bridge Station. Proximate to Dr. Fine Bridge (Hwy 101)

Water Column
Periphyton

County Ramp Station. (On S. side may be above Bailey Hole)

Water Column
Periphyton
Continuous Multiparameter

Above Rowdy Creek Station. (On S. side via Tedson ranch gravel bar)

Water Column
Periphyton
Continuous Multiparameter

Below Rowdy Creek Station. (Above Cattle Crossing)

Water quality assessments at this station must accommodate the variations which occur because of tidal and oceanic influences. Accordingly, the discharger shall employ qualified personnel to obtain samples and data recordings which are representative of the range of conditions encountered (a) between lower-low and higher-high tide levels and (b) between that portion of the water column which is above the pycnocline and the portion which is below the pycnocline.

Monitoring and sampling equipment shall be operated in a manner that concurrently characterizes the "upper band" (low density water, above the pycnocline) and the "lower band" (high density, below the pycnocline) as they ebb and flood past a fixed point.

Water Column (upper and lower)
Periphyton (lower only)
Continuous Multiparameter (upper and lower)

REPORTING

Monitoring reports shall be submitted to the Board for each month on or before the fifteenth day of the following month. In reporting the monitoring data, the discharger shall arrange the data on a form approved by the Executive Officer. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements. Copies of each monitoring report shall be mailed to:

Regional Administrator
U.S. Environmental Protection Agency
Region 9
215 Fremont Street
San Francisco, CA 94105

Ordered by _____

Benjamin D. Kor
Executive Officer

April 6, 1990